Algebra

Quadratic Functions



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Chapter 1

Completing The Square

In elementary algebra, completing the square is a technique for converting a quadratic polynomial of the form

$$ax^2 + bx + c$$

to the form

$$a(x - h)^2 + k$$

for some values of h and k. Completing the square is used in

- solving quadratic equations,
- deriving the quadratic formula,
- graphing quadratic functions,
- evaluating integrals in calculus, such as Gaussian integrals with a linear term in the exponent,
- finding Laplace transforms.

Given any monic quadratic

$$x^2 + bx + c$$

it is possible to form a square that has the same first two terms:

$$\left(x + \frac{1}{2}b\right)^2 = x^2 + bx + \frac{1}{4}b^2$$

This square differs from the original quadratic only in the value of the constant term. Therefore, we can write

$$x^2 + bx + c = \left(x + \frac{1}{2}b\right)^2 + k$$

where $k = c - \frac{b^2}{4}$. This operation is known as completing the square. For example:

$$x^2 + 6x + 11 = (x+3)^2 + 2$$
$$x^2 + 14x + 30 = (x+7)^2 - 19$$
$$x^2 - 2x + 7 = (x-1)^2 + 6$$

Non-monic case [edit] Given a quadratic polynomial of the form

$$ax^2 + bx + c$$

it is possible to factor out the coefficient a, and then complete the square for the resulting monic polynomial.

1.1 Example

$$\begin{aligned} 3x^2 + 12x + 27 &= 3\left(x^2 + 4x + 9\right) \\ &= 3\left((x+2)^2 + 5\right) \\ &= 3(x+2)^2 + 15 \end{aligned}$$

This allows the writing of any quadratic polynomial in the form

$$a(x-h)^2 + k$$

$$ax^2 + bx + c = a(x-h)^2 + k$$

$$\text{Where } h = \frac{-b}{2a} \quad k = c - ah^2 = c - a \cdot (\frac{-b}{2a})^2 = c - \frac{b^2}{4a}$$

$$3x^2 + 12x + 27 \implies h = \frac{-b}{2a} = \frac{-12}{2(3)} = -2, k = c - ah^2 = 27 - 3(-2)^2 = 15$$

$$\text{Hence, } 3x^2 + 12x + 27 = 3(x+2)^2 + 15$$

Formula [edit] Scalar case [edit] The result of completing the square may be written as a formula. For the general case: [1]

$$ax^2 + bx + c = a(x-h)^2 + k, \quad \text{where} \quad h = -\frac{b}{2a} \quad \text{and} \quad k = c - ah^2 = c - \frac{b^2}{4a}$$